Chi-square Tests

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Where this fits: Pipeline map step 0 (Foundations). See the Course map.

0 Foundations	1 Frame	2 Get data	3 Understand	4 Clean	5 Features	6 Reduce
7 Split	8 Model	9 Evaluate	10 Tune	11 Deploy	12 Test	13 Monitor

- **Builds on:** Joint and marginal probability; Hypothesis testing: null and alternative.
- **Leads to:** Filter consistency.
- **Compare with:** One-sample proportion test.

1. Overview

Key point: A chi-square test compares the counts we observe in each category with the counts we would expect if H_0 were true; the bigger the total mismatch χ^2, the smaller the p-value.

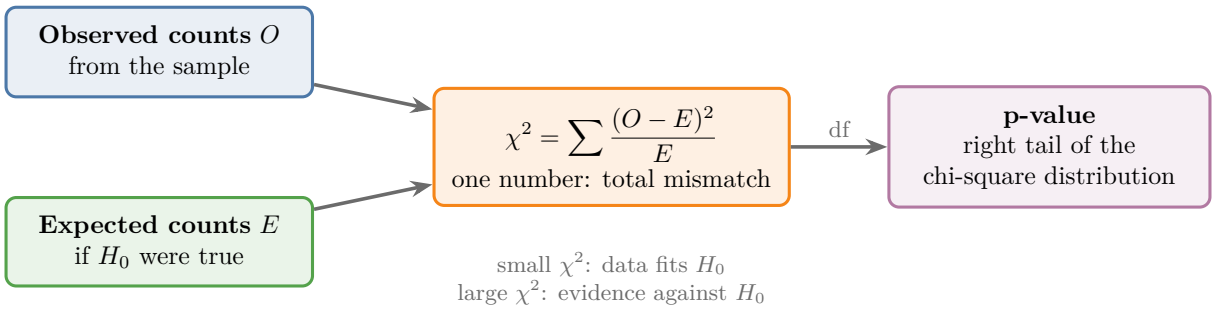


Figure 1: The chi-square test in four steps

The chi-square test was named earlier as the tool of inferential statistics for categorical data. When choosing a hypothesis test, the chi-square test is the test for two categorical **features** (G-772; a feature is a variable of the data, one column of the data table, such as gender). Figure 1 shows how every chi-square test works: observed counts and expected counts go into one number, χ^2, and its tail area under the **chi-square distribution** (G-378) is the p-value.

Two words recur. H_0 is the null hypothesis, the claim that nothing special is going on (for example, that the two features are unrelated). The **p-value** is the probability of a mismatch at least as large as ours if H_0 were true; a small p-value is evidence against H_0 (see one logic for every test).

This Note covers:

- the **chi-square statistic** (G-379) and the chi-square distribution;
- the **goodness-of-fit test**: does one categorical feature match claimed proportions?
- the **test of independence**: are two categorical features related?
- a case study on the Titanic, the conditions the test needs, and its strength measure, **Cramér's V** (G-500).

Extra: All of this Note is extra material beyond the "which test when" guide; the guide only names the chi-square test.

2. The chi-square statistic

Key point: $\chi^2 = \sum(O-E)^2/E$ adds up, over all cells, the squared gap between observed and expected counts, each scaled by the expected count.

2.1 Observed and expected counts

Key point: Observed counts come from the data; expected counts are what H_0 predicts for a sample of the same size.

A categorical feature is summarised by counts, one per category (see the frequency tables). For a chi-square test we need two sets of counts:

- **observed counts** O (G-1375): how many **observations** (G-1374; records, one row of the data table each) of the sample fall in each category or cell;
- **expected counts** E (G-723): how many would fall there, on average, if H_0 were true.

If H_0 is true, O and E differ only by chance. A large mismatch is evidence against H_0.

2.2 The formula

Key point: Square each gap so that positive and negative gaps both count, divide by E so that a gap of 5 matters more in a small cell than in a big one, and add.

1. **In words:** for each cell, take the gap between observed and expected, square it, divide by expected; add over all cells.
2. **Formula:**
$$\chi^2 = \sum_{\text{cells}} \frac{(O-E)^2}{E}$$
3. **Example:** three cells with $O = 20, 26, 14$ and $E = 15, 33, 12$ (section 4):
$$\frac{(20-15)^2}{15} = 1.667$$
$$\frac{(26-33)^2}{33} = 1.485$$
$$\frac{(14-12)^2}{12} = 0.333$$
$$\chi^2 = 1.667 + 1.485 + 0.333 = 3.48$$

χ^2 is 0 only when every observed count equals its expected count. χ^2 can never be negative. This number is the **chi-square statistic** (G-379).

Figure 2 builds it for the example. Each category has its observed and expected bar; the gaps are $+5$, -7 and $+2$. Squaring and dividing by E turns each gap into one block, and the blocks stack into $\chi^2 = 3.48$. The adult gap is the largest, but its block is smaller than the child block, because 7 out of an expected 33 is a smaller surprise than 5 out of an expected 15.

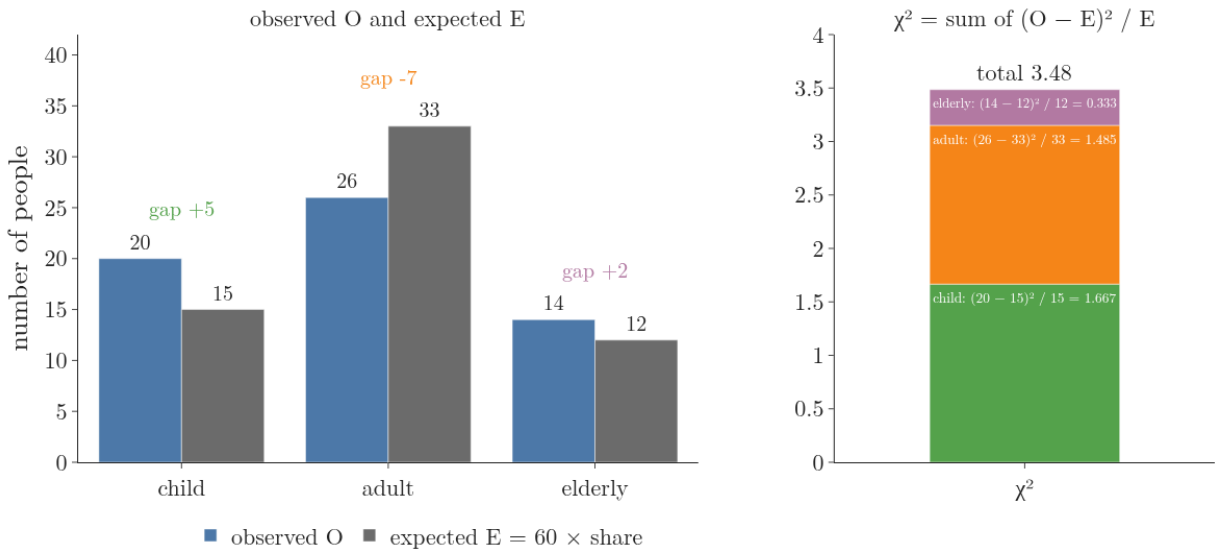


Figure 2: The chi-square statistic built cell by cell for the age groups against the census shares. Left: observed and expected counts, with each gap. Right: each cell's $(O - E)^2 / E$ stacked into the total, 3.48.

3. The chi-square distribution

> **Key point:** If H_0 is true, χ^2 follows a **chi-square distribution** (G-378): never negative, skewed to the right, with one parameter, the degrees of freedom.

3.1 Built from squared normal draws

> **Key point:** Square a draw from the standard normal distribution and we get a chi-square value with 1 degree of freedom; add k squared draws and we get k degrees of freedom.

To judge whether $\chi^2 = 3.48$ is large, we need to know which values χ^2 takes by chance. The curve that answers this is built from the standard normal distribution (see the standard normal) in two steps. Figure 3 runs them.

1. **Square one draw.** Take a number z from the standard normal distribution and square it. Most draws lie between -1 and 1, and squaring a number below 1 makes it smaller still, so most squares pile up next to 0. A square of 4 needs a draw of 2 or -2, which is rare. A square is never negative.

2. **Add several squares.** Add the squares of 2 independent draws, then of 3, then of 5. With 2 squares the curve is still highest at 0. With 3 or more, the sum is rarely near 0, because all the draws would have to be small at once: the peak leaves 0 and the whole curve moves to the right.

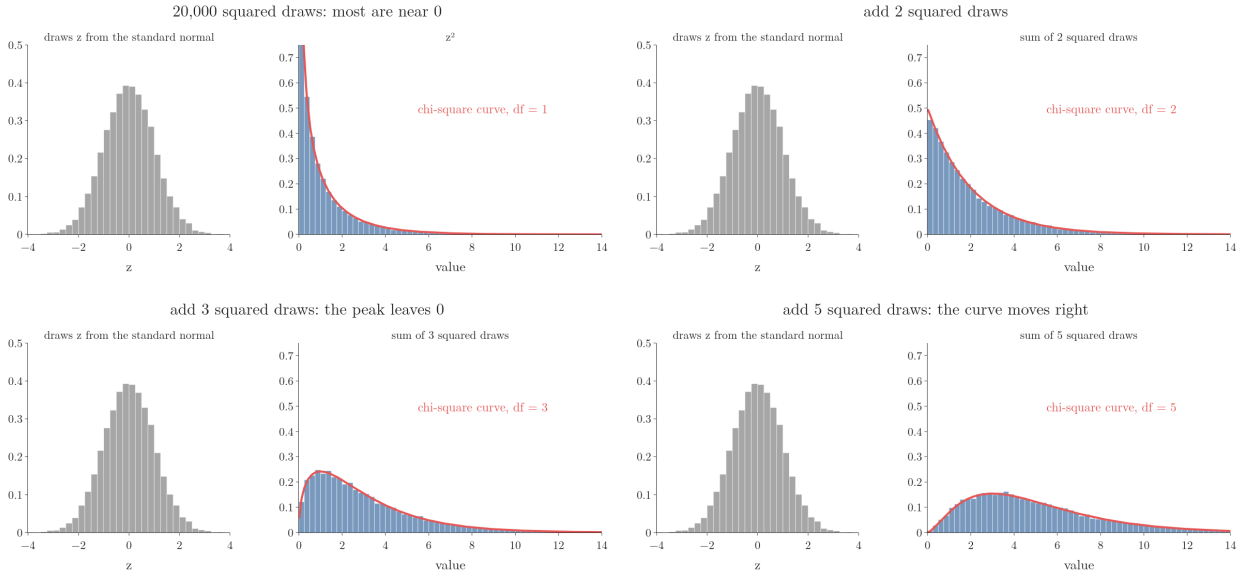


Figure 3: Left: 20,000 draws from the standard normal distribution. Right: the histogram of one squared draw, then of the sum of 2, 3 and 5 squared draws, with the chi-square density of 1, 2, 3 and 5 degrees of freedom (red). Idea after Khan Academy, "Chi-square distribution introduction".

The distribution of a sum of k squared standard normal draws is the **chi-square distribution** (G-378), and k is its **degrees of freedom** (G-578; df). This is its definition.

3.2 The curves and the tail area

> **Key point:** More degrees of freedom move the curve to the right; the p-value is the area to the right of our χ^2.

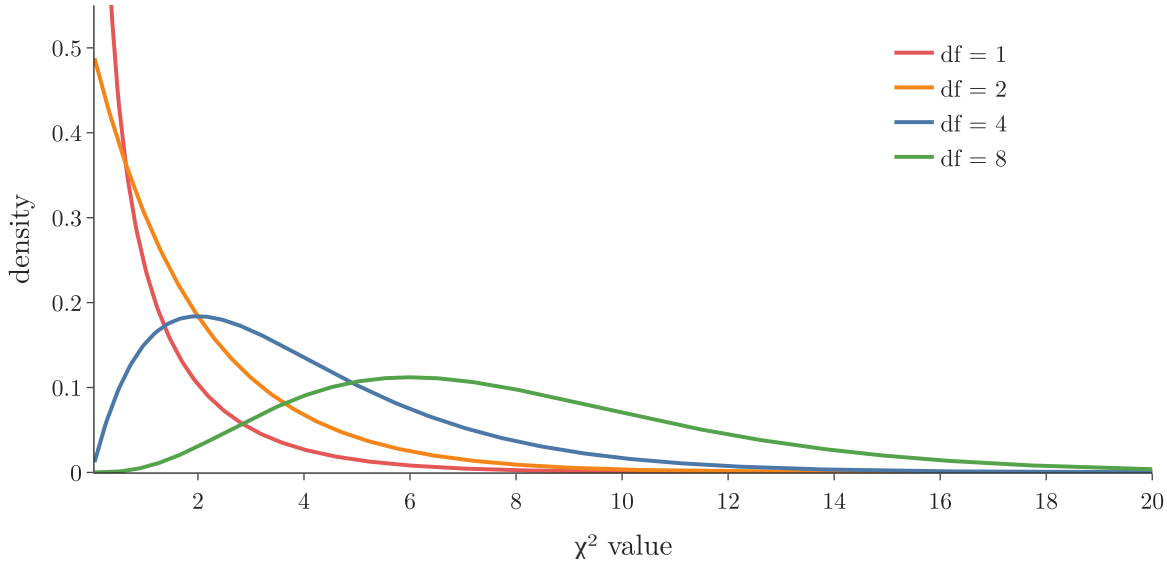


Figure 4: Chi-square densities for four degrees of freedom

The **chi-square distribution** (G-378) is a continuous distribution on the positive numbers. Like Student's t-distribution (see the **t-procedure** (G-1938)), it has one parameter, the degrees of freedom. Figure 4 shows four of them:

- with 1 or 2 df, the density is highest at 0 and falls steadily;
- with more df, the peak moves right and the curve spreads out;
- the mean of a chi-square distribution equals its df (NIST Handbook §1.3.6.6.6).

Only large χ^2 values count as evidence against H_0, so the p-value is always the **right-tail** area beyond our χ^2 (a **right-tailed test**, G-1693). The 5% **critical values** (G-504; the χ^2 above which the p-value falls below 0.05) are 3.84 for 1 df, 5.99 for 2 df and 7.81 for 3 df.

> **Extra:** Why the statistic of section 2 follows this curve. Each $(O - E)/\sqrt{E}$ is roughly normal for large counts, so χ^2 is a sum of squared, roughly normal terms. The terms are not independent, because the counts must add up to n. Pearson (1900) showed that χ^2 still follows a chi-square distribution for large samples, with one df lost to that constraint. The notebook checks this: 200,000 samples of 60 people drawn with the census shares of section 4 true give a mean χ^2 of 2.00, equal to $df = 2$, and 4.9% of them pass the 5% critical value 5.99.

4. The goodness-of-fit test

> **Key point:** One categorical feature against claimed proportions: expected count = sample size × claimed proportion, and the degrees of freedom are the number of categories minus 1.

4.1 The question

> **Key point:** Do the age groups in our sample of 60 match a population that is 25% children, 55% adults and 20% elderly?

The **goodness-of-fit test** (G-853) asks whether the counts of one categorical feature fit a claimed distribution. We use the 60 people of the example dataset. Suppose a census says the population is 25% children, 55% adults and 20% elderly.

- H_0: the age groups occur in the claimed proportions 0.25, 0.55, 0.20;
- H_1: at least one proportion differs.

4.2 Expected counts and the statistic

> **Key point:** each expected count is 60 times the claimed share: 15 children, 33 adults and 12 elderly. Together they give $\chi^2 = 3.48$.

1. **In words:** the expected count of each category is the sample size times its claimed share.
2. **Formula:**

$$E_i = n \times \pi_i$$

$$df = k - 1$$

   where n is the sample size, π_i is the claimed share of category i (here π is a proportion, not 3.14) and k is the number of categories.
3. **Example:**

	child	adult	elderly	total
observed O	20	26	14	60
claimed share π	0.25	0.55	0.20	1
expected $E = 60\pi$	15	33	12	60
$(O-E)^2/E$	1.667	1.485	0.333	$\chi^2 = 3.48$

The degrees of freedom are $k - 1 = 2$: once two expected counts are known, the third is fixed by the total of 60.

4.3 The decision

> **Key point:** $\chi^2 = 3.48$ with 2 df gives $p = 0.175 > 0.05$: we fail to reject H_0.

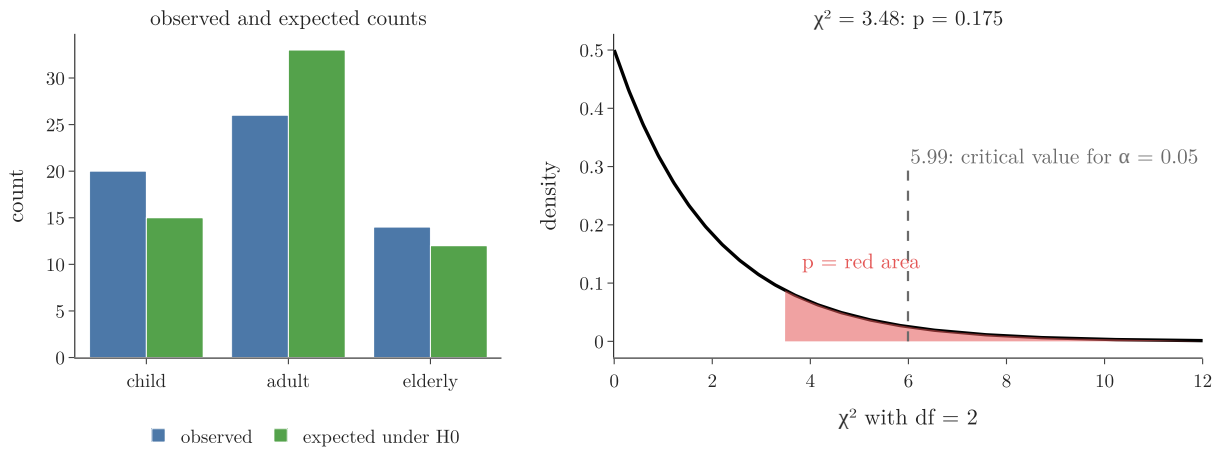


Figure 5: Goodness of fit for the age groups: the counts (left) and the p-value as a tail area (right)

Figure 5 shows both halves of the test. On the left, more children and fewer adults were observed than expected. On the right, $\chi^2 = 3.48$ sits below the critical value 5.99, and the red tail area beyond it is $p = 0.175$.

Since $0.175 > 0.05$, we fail to reject H_0: the sample is consistent with the census proportions. The gaps on the left are within what chance produces in 60 people.

> **Python:** `chisquare` takes the observed and expected counts.
>
> ```
> from scipy import stats
>
> stats.chisquare(f_obs=[20, 26, 14], f_exp=[15, 33, 12])
> # statistic = 3.48, pvalue = 0.175
> ```
>
> Without `f_exp`, scipy assumes equal proportions in all categories.

4.4 Two categories: the proportion test again

> **Key point:** With two categories, the goodness-of-fit χ^2 equals z^2 of the one-sample proportion test, and the p-values are identical.

For gender (26 men, 34 women) against a 50/50 claim, the expected counts are 30 and 30:

$$\frac{(26-30)^2}{30} = 0.533$$

$$\frac{(34-30)^2}{30} = 0.533$$

$$\chi^2 = 0.533 + 0.533 = 1.067$$

With 1 df this gives $p = 0.302$. The **one-sample proportion test** (G-1381) gave $z = -1.033$, and its square is the same number:

$$(-1.033)^2 = 1.067$$

The p-value is the same too, 0.302. The **goodness-of-fit test** (G-853) is the proportion test extended to any number of categories.

> **Extra:** Goodness of fit also checks whether counts follow a named distribution, such as Poisson (see the **Poisson distribution** (G-1509)). The expected counts then come from the PMF, and every parameter estimated from the data (such as λ from the mean) removes one more degree of freedom:
>
> $$df = k - 1 - (\text{number of estimated parameters})$$
>
> The notebook tests this: 5,000 samples of 200 Poisson counts in 5 categories, with λ estimated each time, give a mean χ^2 of 3.03, matching $df = 3$ rather than 4. Judged with the df = 4 critical value, only 2.2% of the samples are rejected instead of 5%; with df = 3 the rate is 5.1%.

5. The test of independence

> **Key point:** Two categorical features in a **contingency table** (G-464): if they are independent, each expected count is row total × column total / grand total, and $df = (r-1)(c-1)$.

5.1 The question and the table

> **Key point:** Is the gender mix the same in every age group, or do gender and age group go together?

The **chi-square test of independence** (G-380) asks whether two categorical features are related. The counts go in a **contingency table** (G-464; see the **contingency tables** (G-464)):

	child	adult	elderly	row total
female	12	12	10	34
male	8	14	4	26
column total	20	26	14	60

- H_0: gender and age group are independent;
- H_1: they are related.

5.2 Expected counts under independence

> **Key point:** Independence means $P(A \text{ and } B) = P(A)\,P(B)$; multiplied by n, this gives $E = \text{row total} \times \text{column total}/n$.

Start from the plain idea. Of the 60 people, 34 are female: a share of 56.7 percent:

$$\frac{34}{60} = 0.567$$

If age group has nothing to do with gender, every age group should have that same share of women. So of the 20 children we expect this many girls:

$$20 \times 0.567 = 11.33$$

And of the 26 adults we expect this many women:

$$26 \times 0.567 = 14.73$$

The same rule in the language of probability: two events are independent when the probability of both is the product of their probabilities (see the **independent events** (G-934)). The **marginal probabilities** (G-1165) come from the totals (see the joint and marginal probability).

1. **In words:** the expected count of a cell is its row total times its column total, divided by the grand total.
2. **Formula:**
$$E = n \times \frac{\text{row total}}{n} \times \frac{\text{column total}}{n}$$
$$E = \frac{\text{row total} \times \text{column total}}{n}$$
3. **Example:** female children. The two probabilities are:
$$P(\text{female}) = \frac{34}{60}$$
$$P(\text{child}) = \frac{20}{60}$$
So the expected count is:
$$E = \frac{34 \times 20}{60}$$
$$E = \frac{680}{60} = 11.33$$

All six expected counts:

	child	adult	elderly	row total
female	11.33	14.73	7.93	34
male	8.67	11.27	6.07	26
column total	20	26	14	60

The expected table has the same totals as the observed one; only the inside is spread out "as if independent".

Figure 6 sets the observed and expected tables side by side, with the third table holding each cell's contribution to χ^2, computed in the next section. The darkest contribution cells are the elderly men (4 observed, 6.07 expected) and the adult men (14 observed, 11.27 expected).

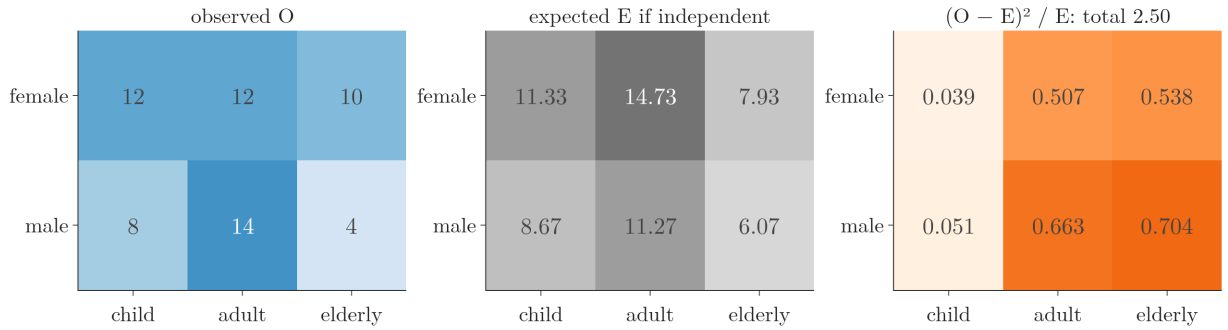


Figure 6: The test of independence for gender by age group. Left: observed counts. Middle: counts expected if gender and age group were independent, row total × column total / 60. Right: each cell's $(O - E)^2 / E$; the six add up to $\chi^2 = 2.50$.

5.3 The statistic and the decision

> **Key point:** $\chi^2 = 2.50$ with 2 df gives $p = 0.29$: no evidence that gender and age group are related.

The contribution $(O - E)^2/E$ of each cell; for female children it is:

$$\frac{(12 - 11.33)^2}{11.33} = 0.039$$

All six:

	child	adult	elderly
female	0.039	0.507	0.538
male	0.051	0.663	0.704

1. **In words:** add the six contributions; the degrees of freedom are (rows − 1) times (columns − 1).

2. **Formula:**

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$
$$df = (r-1)(c-1)$$

3. **Example:**

$$\chi^2 = 0.039 + 0.507 + 0.538$$
$$+ 0.051 + 0.663 + 0.704$$
$$\chi^2 = 2.50$$
$$df = (2-1)(3-1) = 2$$

With 2 df, the p-value is:

$$p = P(\chi^2 \geq 2.50) = 0.29$$

Since $0.29 > 0.05$, we fail to reject H_0. The elderly group has relatively few men (4 observed, 6.07 expected), but in a sample of 60 a gap like this is common by chance.

Why $(r-1)(c-1)$: with the row and column totals fixed, filling $(r-1)(c-1)$ cells freely determines all the others. In a 2 by 3 table, choosing 2 cells of the female row fixes the rest.

Python: `chi2_contingency` takes the observed table and returns the statistic, p-value, df and expected counts.

```
import pandas as pd

table = pd.crosstab(people["gender"], people["age_group"])
result = stats.chi2_contingency(table)
result.statistic, result.pvalue, result.dof    # 2.50, 0.29, 2
result.expected_freq                           # the expected table
```

6. Case study: survival on the Titanic

Key point: Sex and class were both strongly related to survival: $\chi^2 = 263$ ($p \approx 10^{-59}$) for sex, $\chi^2 = 103$ ($p \approx 10^{-23}$) for class.

6.1 Sex and survival

Key point: 81 women died where independence predicts 194; 109 men survived where it predicts 222.

The Kaggle Titanic training file has 891 passengers. Survival against sex:

	died	survived
female, observed	81	233

	died	survived
female, expected	193.5	120.5
male, observed	468	109
male, expected	355.5	221.5

Each cell contributes (observed minus expected) squared, divided by expected:

Cell	Work	Contribution
female, died	$(81 - 193.5)^2/193.5$	65.4
female, survived	$(233 - 120.5)^2/120.5$	105.0
male, died	$(468 - 355.5)^2/355.5$	35.6
male, survived	$(109 - 221.5)^2/221.5$	57.1

The four contributions add up to $\chi^2 = 263.1$ with $df = 1$, and the p-value is tiny:

$$p = 3.7 \times 10^{-59}$$

We reject H_0: survival depended on sex.

> **Extra:** For a 2 by 2 table, `chi2_contingency` applies **Yates' continuity correction** (G-2135) by default: it shrinks each $|O - E|$ by 0.5 before squaring (Yates 1934), which lowers χ^2 and so makes the test slightly more cautious. Here it gives $\chi^2 = 260.7$ instead of 263.1; pass `correction=False` to get the plain statistic. With counts this large the choice does not matter.

6.2 Class and survival

> **Key point:** First class had 136 survivors where independence predicts 83; third class had 119 where it predicts 188.

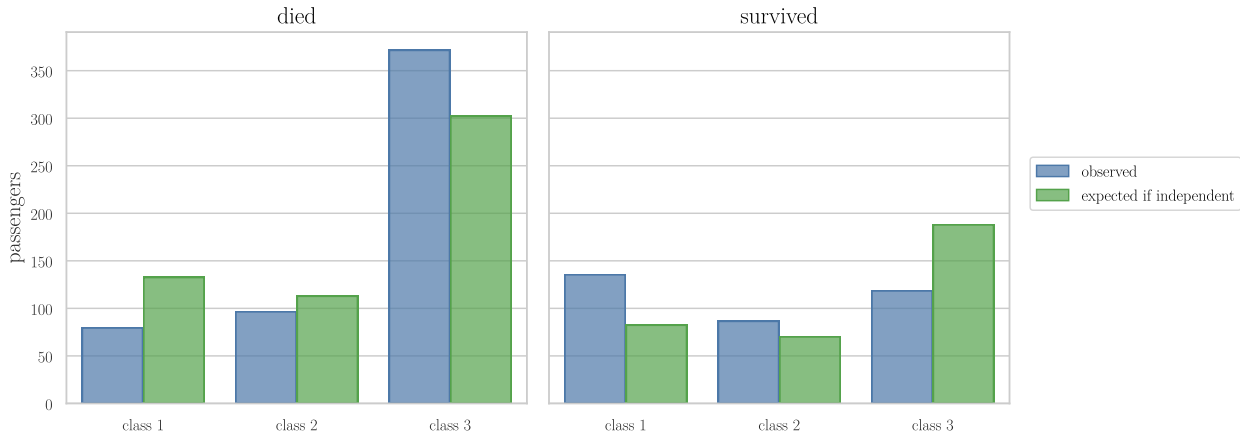


Figure 7: Titanic passengers by class and outcome: observed counts and counts expected under independence

The crosstab of class against survival was drawn as a heatmap in the bivariate analysis. Figure 7 compares its observed counts with the expected ones. First class has far more survivors than independence predicts, third class far fewer. The survival rates were 63%, 47% and 24% for the three classes, against 38% overall.

The test gives $\chi^2 = 102.9$ with 2 degrees of freedom:

$$df = (3 - 1)(2 - 1) = 2$$

The p-value is tiny:

$$p = 4.5 \times 10^{-23}$$

We reject H_0: survival depended on class.

7. Conditions and limits

> **Key point:** Use raw counts, independent observations, and expected counts of at least 5 in every cell; a significant χ^2 says that a relationship exists, not how strong it is.

7.1 Conditions

> **Key point:** Counts, independence, and large enough expected counts.

1. **Counts, not percentages.** The formula needs the number of observations in each cell. Percentages or means give a wrong χ^2, because χ^2 grows with the number of observations (section 7.2): percentages act as if every sample had exactly 100 observations, whatever its real size.

2. **Independent observations.** Each observation is counted in exactly one cell, and observations do not influence each other. The same person measured twice breaks this.
3. **Large enough expected counts.** The usual rule is that every expected count should be at least 5 (Cochran 1954), because the **chi-square distribution** (G-378) is only a large-sample approximation. All six expected counts of section 5.2 are above 5 (the smallest is 6.07). For small 2 by 2 tables, **Fisher's exact test** (G-782) (`stats.fisher_exact`) avoids the approximation; for bigger tables, merging rare categories helps.

Extra: What goes wrong below the rule. A ruler marked in metres is fine for a room but useless for a grain of rice; the chi-square curve is likewise only a good ruler for large counts. The notebook draws 200,000 samples with H_0 true, so a correct 5% test should reject 5% of them:

Claimed shares	n	Smallest E	Rejected at 5%
0.25, 0.55, 0.20	60	12	4.9%
0.90, 0.05, 0.05	8	0.4	11.2%
0.90, 0.05, 0.05	10	0.5	2.7%

With large expected counts the test keeps its 5% promise. With two rare categories and tiny samples, the false-alarm rate drifts away from 5%, more than double in one case and about half in the other, so the p-value can no longer be trusted.

Figure 8 shows why. With large expected counts the simulated χ^2 values spread smoothly along the chi-square curve. With expected counts below 1, only a few count patterns are possible, so χ^2 can take only a few values: it piles up on a handful of spikes that the smooth curve cannot describe, and the share beyond the 5% line (dashed) is wrong.

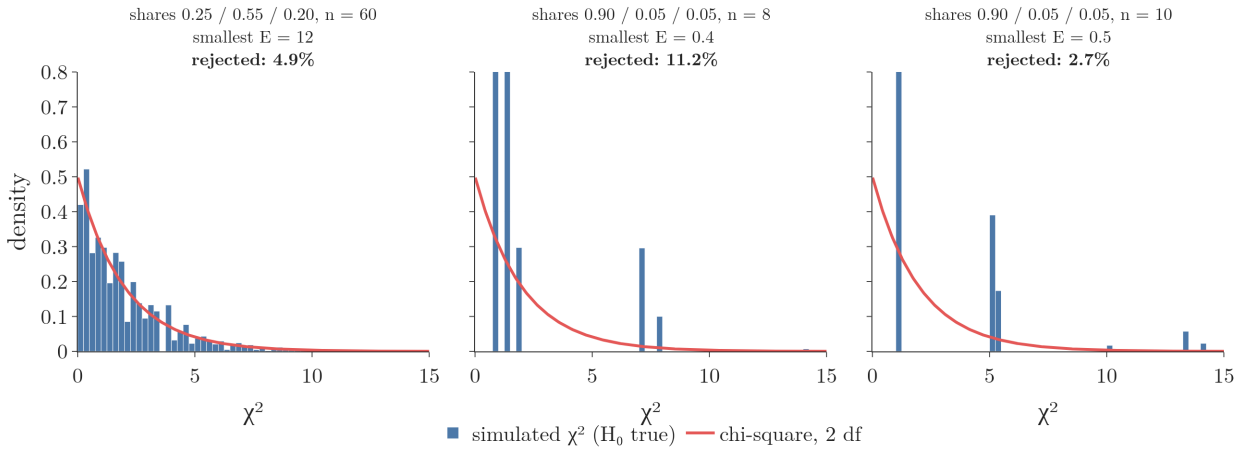


Figure 8: Simulated χ^2 values with H_0 true, 200,000 samples each (bars), against the chi-square distribution with 2 df (curve). Left: every expected count is at least 12, and the curve fits. Middle and right: expected counts of 0.4 and 0.5 make χ^2 pile up on a few values, and the share beyond the 5% critical value (dashed) becomes 11.2% or 2.7%.

7.2 Strength of a relationship: Cramér's V

> **Key point: Cramér's V** (G-500) rescales χ^2 to a number between 0 (no relationship) and 1 (perfect relationship) (Cramér 1946).

χ^2 grows with the sample size: the same pattern in 10 times as many observations gives a 10 times larger χ^2. To measure strength, we divide the size out.

1. **In words:** divide χ^2 by the sample size and by one less than the smaller table dimension, then take the square root.
2. **Formula:**

$$V = \sqrt{\frac{\chi^2}{n \times (\min(r, c) - 1)}}$$

   Here $\min(r, c)$ is the smaller of the number of rows r and columns c.
3. **Example:** Titanic sex and survival, $\chi^2 = 263.1$, $n = 891$, a 2 by 2 table:

$$\frac{263.1}{891 \times 1} = 0.295$$

$$V = \sqrt{0.295} = 0.54$$

For class and survival:

$$V = \sqrt{\frac{102.9}{891}} = 0.34$$

Both relationships are significant, but sex was the stronger predictor of survival. In scipy: `stats.contingency.association(table, method="cramer")`.

Figure 9 shows why the rescaling is needed. The Titanic sex-by-survival table is shrunk to a tenth and grown to ten times its size, keeping the same pattern. χ^2 grows with the number of passengers, from 26.3 to 263.1 to 2,630.5, but V stays at 0.54 every time. (The blue χ^2 bars use a log scale: each step up the axis multiplies the value by 10.)

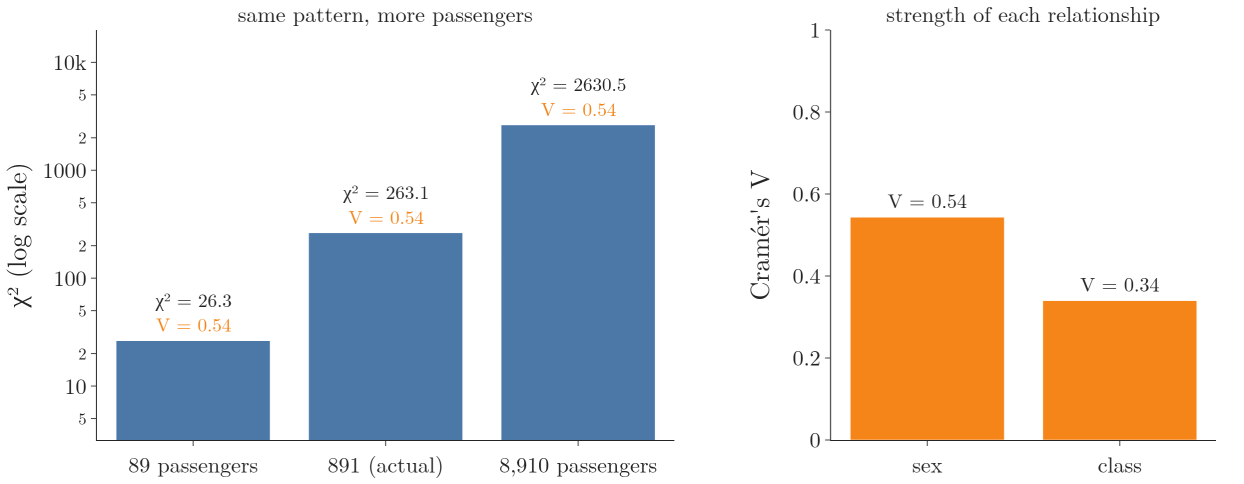


Figure 9: Left: the Titanic sex-by-survival pattern with a tenth, the actual and ten times the passengers. χ^2 (blue, log scale) grows with the sample; Cramér's V stays 0.54. Right: V for sex (0.54) and for class (0.34) against survival.

7.3 Chi-square in machine learning

> **Key point:** Feature selection with `SelectKBest(chi2)` ranks categorical features by how strongly they are related to the target.

Feature selection (G-768) with **`SelectKBest`** (G-1762; keeps the k features with the highest scores) **`(score_func=chi2)`** (see the **pipelines** (G-1499)) scores each feature against the **target** (G-1949; the output we predict) with a **chi-square statistic** (G-379): a feature whose counts differ strongly between the classes gets a high score. scikit-learn's version treats each feature's values as counts, which is why it needs values of 0 or more (scikit-learn docs, `chi2`). The score is not the same number as the test of independence. For a one-hot column it compares only the rows where the column is 1. In the notebook, Titanic sex one-hot encoded gets the scores 170.3 (female) and 92.7 (male); only their sum, 263.1, equals the χ^2 of section 6.1.

8. Summary

	Goodness of fit	Test of independence
Features	one categorical	two categorical
H_0	the categories have the claimed proportions	the two features are independent
Expected count	$n \times \pi_i$	row total $\times$ column total $/n$
Degrees of freedom	$k - 1$	$(r-1)(c-1)$
scipy	`chisquare`	`chi2_contingency`
Example here	age groups vs census: $\chi^2 = 3.48$, $p = 0.175$	gender vs age group: $\chi^2 = 2.50$, $p = 0.29$

- The table's expected counts are what H_0 predicts: n times the claimed share, or, under independence, $P(A)\,P(B)$ times n. Both examples give $p > 0.05$, so neither sample shows a departure from H_0.
- Under H_0, $\chi^2 = \sum(O-E)^2/E$ follows a **chi-square distribution** (G-378); the p-value is the right-tail area, because only a large total mismatch between observed and expected counts counts against H_0.
- With two categories, goodness of fit is the **one-sample proportion test** (G-1381): $\chi^2 = z^2$, so the two tests give identical p-values.
- The test needs raw counts, independent observations and expected counts of at least 5, because χ^2 grows with the number of observations, so percentages give a wrong χ^2, and with tiny expected counts the chi-square curve no longer fits, so the p-value cannot be trusted.
- This answers the opening question: a chi-square test compares observed counts with the counts H_0 predicts, and the bigger the total mismatch χ^2, the smaller the p-value.

9. Sources

Built from

- Krish Naik, "Tutorial 32- All About P Value,T test,Chi Square Test, Anova Test and When to Use What?", YouTube, https://www.youtube.com/watch?v=YrhlQB3mQFI
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- Khan Academy, "Pearson's chi square test (goodness of fit) | Probability and Statistics | Khan Academy", YouTube, https://www.youtube.com/watch?v=2QeDRsxSF9M
- Khan Academy, "Contingency table chi-square test | Probability and Statistics | Khan Academy", YouTube, https://www.youtube.com/watch?v=hpWdDmgsIRE

Other references

- Cochran, W. G. (1954). "Some Methods for Strengthening the Common χ^2 Tests". *Biometrics* 10(4).
- Cramér, H. (1946). *Mathematical Methods of Statistics*. Princeton University Press.

- NIST/SEMATECH (2012). *e-Handbook of Statistical Methods*. itl.nist.gov/div898/handbook. Section 1.3.6.6.6, chi-square distribution.
- Pearson, K. (1900). "On the criterion that a given system of deviations from the probable in the case of a correlated system of variables is such that it can be reasonably supposed to have arisen from random sampling". *Philosophical Magazine* 50(302).
- scikit-learn documentation. `sklearn.feature_selection.chi2`.
- Yates, F. (1934). "Contingency Tables Involving Small Numbers and the χ^2 Test". *Supplement to the Journal of the Royal Statistical Society* 1(2).

10. Key terms

Terms taught in this Note come first; linked terms are recaps, taught in the Note the link opens.

Term	Meaning
Chi-square distribution (G-378)	The shape the chi-square statistic χ^2 follows when the null hypothesis H_0 is true: never negative, skewed to the right, with one parameter, the degrees of freedom.
Chi-square statistic χ^2 (G-379)	A number that measures the total mismatch between observed and expected counts, $\sum(O-E)^2/E$; it is 0 when every count matches and grows as they differ.
Cramér's V (G-500)	A measure of how strongly two categorical features are related, from 0 (no relationship) to 1 (perfect relationship), made by rescaling the chi-square statistic.
Observed count O (G-1375)	The number of sample rows in a category or table cell.
Expected count E (G-723)	The number of rows a category or cell would hold on average if the null hypothesis H_0 were true.
Right-tailed and left-tailed test (G-1693)	One-tailed tests that count only one tail as evidence against H_0: a right-tailed test rejects for large values of the test statistic (as for $H_1 : \mu > \mu_0$), a left-tailed test for small ones (as for $H_1 : \mu < \mu_0$).
Goodness-of-fit test (G-853)	A check of whether the counts in one categorical column match the shares we claimed (a chi-square test against claimed proportions); $df = k - 1$ for k categories.
Chi-square test of independence (G-380)	A chi-square test of whether two categorical columns are related; $df = (r-1)(c-1)$.

Term	Meaning
Yates' continuity correction (G-2135)	A small adjustment to χ^2 for 2 by 2 tables, applied by default in `chi2_contingency`.
Fisher's exact test (G-782)	A test for small 2 by 2 count tables that computes the p-value exactly instead of approximating it; used when expected counts fall below 5.
Chi-square test (G-381)	A test of whether categorical counts match expected counts (goodness of fit) or whether two categorical columns are related (independence).
Feature (G-772)	One piece of information about each example that a model uses (e.g. a student's CGPA).
Observation (G-1374)	One record of a dataset: one row of the data table.
Degrees of freedom (G-578)	The number of deviations free to vary, $n - 1$ for a sample of size n; it sets the shape of the t-distribution (its parameter).
T-procedure (G-1938)	The way to build a confidence interval for a mean when the population standard deviation σ is unknown: use the sample's s and a t critical value in place of z, giving $\bar{x} \pm t_{\alpha/2,\, n-1}\, s/\sqrt{n}$.
Critical value (G-504)	The cut-off on a z (or t) curve that leaves $\alpha/2$ in each tail; a confidence interval reaches this many standard errors either side of the estimate. It is 1.96 for 95% on the standard normal curve.
One-sample proportion test (G-1381)	A test of whether the share (proportion) of one category in the population equals a claimed value π_0, using a z statistic (a z-test).
Poisson distribution (G-1509)	A distribution that gives the probability of each possible count of events (0, 1, 2, ...) in a fixed interval of time or space, from the average count λ; used for counts such as questions per day.
Contingency table (G-464)	A table that counts the observations for every pair of categories of two categorical features, so we can see whether the two are related; also called a crosstab.

Term	Meaning
Independent events (G-934)	Events where one happening does not change the probability of the other; for them the probability that both happen is the product of their probabilities.
Marginal probability distribution (G-1165)	All the marginal probabilities of one variable, read from a joint table.
Feature selection (G-768)	Keeping only the useful input columns and dropping the rest.
SelectKBest (G-1762)	scikit-learn class that gives every column a score and keeps the `k` best, so only the most useful features go into the model.
Pipeline (G-1499)	One object that chains preprocessing steps and a model, each step's output feeding the next; every input, including new ones after deployment, then gets exactly the same preprocessing.
Target (G-1949)	The output column we predict, such as the class; also called the label.